Sample Final Exam: Questions and One-Paragraph Answers

DATA 642 – Statistical Machine Learning

Module 1 — Foundations

1. Explain how geometric intuition can be used to understand the solution behavior of least squares regression. Discuss when geometric intuition breaks down in high-dimensional spaces. **Answer:** Least squares regression can be understood geometrically by noting that the fitted values represent the orthogonal projection of the response vector onto the column space of the design matrix. This provides intuition for residuals, orthogonality conditions, and the uniqueness of the solution. However, this intuition breaks down in high-dimensional settings because geometric relationships such as angles, distances, and orthogonality behave counterintuitively—most vectors become nearly orthogonal, volume becomes sparse, and projection interpretation becomes less meaningful due to concentration of measure effects.

2. Describe the role of orthogonality in optimization algorithms. **Answer:** Orthogonality ensures that updates do not interfere destructively with each other and often simplifies convergence analysis. For example, in conjugate gradient methods, updates are constructed to be orthogonal with respect to a matrix-induced inner product, reducing redundancy between search directions. Orthogonality also helps interpret gradient descent behavior when gradients point consistently in directions that reduce the objective. However, in non-convex settings or ill-conditioned problems, orthogonality is difficult to maintain and becomes less informative.

3. Discuss how the conditioning of a matrix influences gradient-based optimization. **Answer:** Poor conditioning means the matrix has eigenvalues with large disparity, stretching level sets of the objective into elongated ellipsoids. Gradient descent must take very small steps along directions of high curvature, slowing convergence. This manifests in zig-zagging trajectories and unstable progress. Well-conditioned problems have more spherical level sets, allowing large, reliable steps. Consequently, preconditioning and regularization techniques are crucial for improving numerical stability and convergence speed.

4. Explain why convexity guarantees a unique minimizer but does not necessarily guarantee fast convergence. **Answer:** Convexity ensures that any local minimizer is global, and strict convexity ensures uniqueness. However, the speed of convergence depends on curvature properties, such as Lipschitz continuity of the gradient and strong convexity. Flat regions or poorly conditioned curvature slow down gradient-based methods even though a unique minimum exists. Hence convexity guarantees correctness of the solution, not efficiency of reaching it.

5. Compare constrained and unconstrained optimization from a conceptual perspective. **Answer:** Unconstrained optimization seeks a minimum in the full space, whereas constrained optimization restricts the feasible region, often forcing solutions to lie on boundaries where gradients vanish only after accounting for constraints. Constraint sets can add nonlinearity or nonconvexity and require projection or Lagrangian-based techniques. In many practical cases—such as physical systems or resource-limited problems—constraints fundamentally alter both the feasible set and the nature of optimality.

6. Provide intuition for why gradient descent may converge to saddle points in high-dimensional problems. **Answer:** Saddle points are abundant in high dimensions because random directions are far more likely to include both positive and negative curvature than purely negative curvature. Gradient descent, which relies solely on first-order information, cannot distinguish shallow minima from saddle points when gradients are small. Without curvature information, it may stagnate or proceed very slowly. While noise or momentum can help escape, saddle points remain a fundamental difficulty in non-convex optimization.

7. Describe the use of projection operators in optimization and what happens when the feasible set is non-convex. **Answer:** Projection operators enforce feasibility by mapping iterates to the closest point in a constraint set. In convex sets, projection is uniquely defined and preserves convergence guarantees for projected gradient methods. In non-convex settings, projections may be non-unique or computationally expensive, and projected gradient descent can oscillate or converge to poor local minima. Thus, projection is far less reliable when constraints create disconnected feasible regions.

8. Explain the difference between strict and strong convexity. **Answer:** Strict convexity ensures that the objective lies strictly above the line segment connecting any two distinct points, guaranteeing a unique minimizer. Strong convexity additionally requires curvature to exceed a positive threshold, implying quadratic growth away from the minimizer. Strong convexity yields faster, more stable convergence for gradient-based algorithms, while strict convexity alone provides uniqueness without performance guarantees.

9. Discuss the meaning of sensitivity in optimization. **Answer:** Sensitivity measures how small perturbations in data or parameters affect the optimal solution. Problems with high sensitivity can exhibit dramatic changes from minor input variations, leading to instability and unreliable predictions. Regularization, conditioning, and robust optimization frameworks are often employed to reduce sensitivity, especially in ill-posed or high-dimensional settings.

10. Explain implicit regularization and how it appears in optimization algorithms. **Answer:** Implicit regularization refers to the tendency of certain algorithms to prefer particular types of solutions even without explicit penalties. For example, gradient descent in linear regression tends to produce minimum-norm solutions. This arises from the geometry of the updates and the structure of the parameter space. Implicit regularization helps explain why overparameterized models can generalize well despite lacking explicit constraints.

Module 2 — Regression, Classification, Regularization

11. Compare geometric interpretations of ridge and lasso regression. **Answer:** Ridge regression imposes an ℓ_2-ball constraint, producing circular or spherical level sets that encourage small but non-sparse solutions. Lasso imposes an ℓ_1-ball, whose corners correspond to sparse solutions, making it more likely to set coefficients exactly to zero. Thus, the geometry of the constraint shapes determines whether sparsity arises naturally.

12. Explain the bias–variance tradeoff in high-dimensional regression. **Answer:** In high dimensions, adding features initially reduces bias but eventually increases variance as the model begins fitting noise rather than signal. This creates a U-shaped test error curve. Regularization counteracts this by reducing variance at the cost of bias. High-dimensionality exaggerates variance, making careful model selection essential.

13. Discuss the role of slack variables in SVMs. **Answer:** Slack variables allow margin violations and enable SVMs to handle noisy or non-linearly separable data. They quantify how much a sample violates the margin and are penalized in the objective, balancing robustness with flexibility. Without slack variables, SVMs would fail on even mildly overlapping data.

14. Explain why logistic regression is considered a discriminative model. **Answer:** Logistic regression models $P(y|x)$ directly, focusing on how the label depends on the input. It does not attempt to model the distribution of x itself. Discriminative modeling typically yields more flexible boundaries and often performs better when class-conditional densities are complex.

15. Provide a conceptual explanation of the kernel trick. **Answer:** The kernel trick implicitly maps data into a higher-dimensional feature space where linear methods become expressive enough to capture nonlinear boundaries. Instead of explicitly constructing the mapping, kernels compute inner products in the feature space, enabling efficient computation and richer function classes.

16. Describe decision boundaries in high-dimensional SVMs. **Answer:** In high dimensions, SVM decision boundaries remain hyperplanes but can exploit a vast array of orthogonal directions to separate classes. Margins tend to widen due to increased flexibility, but overfitting risk rises unless regularization is applied. The geometric simplicity of hyperplanes belies their expressive power in high dimensions.

17. Explain how feature scaling affects optimization in regression. **Answer:** Scaling reduces discrepancies in feature magnitude, preventing gradients associated with large-scale features from dominating updates. This stabilizes optimization and ensures that learning rates apply uniformly across coefficients. Unscaled features often cause slow convergence and numerical instability.

18. Describe the impact of Tikhonov regularization on numerical stability. **Answer:** Ridge regression adds a multiple of the identity matrix to the covariance matrix, improving its conditioning and making matrix inversion more stable. This mitigates issues arising from multicollinearity or near-singularity, ensuring robust parameter estimates.

19. Explain how stochastic optimization influences logistic regression. **Answer:** Stochastic updates introduce randomness that helps explore parameter space more broadly and avoid poor local minima in non-convex variants. They also reduce per-iteration cost and act as a form of implicit regularization. The resulting model can generalize better despite noisier updates.

20. Discuss why overparameterized models can still generalize well. **Answer:** Overparameterized models often learn simple patterns before fitting noise, a phenomenon tied to implicit bias of optimization algorithms. Additionally, high-dimensional structure can align with meaningful low-dimensional manifolds in the data. This allows good generalization even when classical theory predicts overfitting.

Module 3 — PCA, ICA, Dimensionality Reduction

21. Explain why PCA finds directions of maximal variance. **Answer:** PCA identifies directions that capture the most variability by maximizing the variance of projected data. This corresponds to selecting eigenvectors of the covariance matrix with largest eigenvalues. However, PCA performs poorly when variance does not correspond to meaningful structure, such as when noise dominates signal.

22. Discuss how whitening prepares data for ICA. **Answer:** Whitening removes correlations and standardizes variance in all directions, ensuring that ICA focuses solely on higher-order statistical dependencies such as non-Gaussianity. This preprocessing step simplifies the ICA optimization landscape and helps isolate independent components.

23. Compare PCA and ICA. **Answer:** PCA identifies orthogonal directions of maximal variance, while ICA finds statistically independent components by exploiting non-Gaussianity. PCA decorrelates data, but ICA goes further by removing all linear dependencies. ICA is preferred when latent sources are non-Gaussian or mixed.

24. Explain the meaning of identifiability in ICA. **Answer:** ICA solutions are unique only up to scaling and permutation because multiplying a component by a constant and dividing the mixing vector by the same constant produces identical mixtures. This inherent ambiguity reflects the limitations of the model and does not affect recovery of the underlying independent signals.

25. Describe how low-rank approximations reduce noise. **Answer:** Low-rank approximations remove components associated with small singular values, which typically represent noise rather than meaningful structure. By reconstructing data using only dominant singular vectors, the approximation preserves essential patterns while filtering out irrelevant perturbations.

26. Explain how singular values reflect dataset structure. **Answer:** Singular values quantify the importance of corresponding singular vectors in representing the data. Large singular values signal dominant directions of variation, while small ones signify redundancy or noise. A rapid decay indicates that the dataset lies close to a low-dimensional manifold.

27. Discuss reconstruction error in PCA. **Answer:** Reconstruction error measures the information lost by projecting data onto a lower-dimensional subspace. It corresponds to the sum of omitted eigenvalues. Practically, it indicates how well the chosen components capture the original structure and informs the tradeoff between compression and fidelity.

28. Explain why PCA fails when variables differ in scale. **Answer:** PCA is sensitive to variance, so variables with larger scales disproportionately influence the principal components. Without standardization, PCA may capture measurement units rather than meaningful patterns, leading to distorted representations.

29. Discuss how dimensionality reduction affects cluster structure. **Answer:** Dimensionality reduction can preserve or distort cluster relationships depending on whether reduced components emphasize inter-cluster separation. If important discriminative features lie in low-variance directions, PCA-like techniques may collapse distinct clusters. Nonlinear methods sometimes mitigate this.

30. Explain subspace learning in modern ML. **Answer:** Subspace learning seeks low-dimensional structures that capture essential variation in data. Linear methods like PCA derive global subspaces, while nonlinear approaches uncover curved manifolds. Subspace learning underlies many modern techniques, including autoencoders and manifold learning.

Module 4 — Clustering & Unsupervised Learning

31. Compare k-means and hierarchical clustering. **Answer:** K-means partitions data into clusters by minimizing within-cluster variance, whereas hierarchical clustering builds nested partitions using iterative merging or splitting. Hierarchical methods reveal multi-scale structure and do not require choosing the number of clusters in advance, making them more informative in exploratory analysis.

32. Explain how distance metrics influence clustering outcomes. **Answer:** Distance metrics define similarity and determine cluster shapes. Small metric changes can alter cluster assignments dramatically, especially in high-dimensional data where distances concentrate. Choosing an appropriate metric is essential for meaningful clustering.

33. Provide intuition for why k-means converges despite non-convexity. **Answer:** Each iteration of k-means decreases the objective by alternately assigning points to their nearest centroid and updating centroids. This monotonic decrease guarantees convergence to a local minimum even though global optimality is not assured. The algorithm's alternating structure ensures progress.

34. Describe cluster stability. **Answer:** Cluster stability measures whether clustering results remain consistent under perturbations such as resampling or noise. High stability suggests that detected clusters reflect genuine structure rather than artifacts. It is a key diagnostic tool in unsupervised learning.

35. Explain density-based clustering. **Answer:** Density-based methods identify clusters as contiguous regions of high point density separated by sparse regions. They detect clusters of arbitrary shapes and filter out noise automatically. These approaches excel when data contain irregular clusters not captured by centroid-based methods.

36. Discuss the impact of initialization on clustering algorithms. **Answer:** Initialization determines which basin of attraction the algorithm converges to. Poor initial centroids can trap k-means in suboptimal partitions. Careful seeding strategies like k-means++ mitigate this by spreading initial centroids.

37. Compare top-down and bottom-up clustering strategies. **Answer:** Top-down (divisive) methods recursively split clusters, while bottom-up (agglomerative) methods merge smaller clusters. Bottom-up methods are more common due to computational simplicity. Each strategy produces different dendrogram structures and insights.

38. Explain how clustering assists supervised learning. **Answer:** Clustering can reveal latent groups that serve as features, initialize models, or reduce dimensionality. It also helps identify subpopulations that improve model interpretability or performance in downstream tasks.

39. Discuss ambiguity in clustering. **Answer:** Without assumptions on cluster shape, size, or separation, clustering has no unique solution. Different algorithms or metrics yield different partitions, reflecting the problem's inherent ill-posedness.

40. Describe the interplay between dimensionality reduction and clustering. **Answer:** Dimensionality reduction can simplify clusters and mitigate noise but may distort separation if important discriminative features lie in discarded directions. Thus, combining the two requires careful validation.

Module 5 — Neural Networks, Autoencoders, Deep Learning

41. Explain how a neural network represents nonlinear functions. **Answer:** Neural networks compose linear transformations with nonlinear activations, creating hierarchical representations capable of approximating complex functions. Depth allows these compositions to capture intricate patterns that shallow models struggle with, giving networks universal approximation power.

42. Discuss the intuition behind backpropagation. **Answer:** Backpropagation applies the chain rule across the network's computation graph, efficiently computing gradients by reusing intermediate derivatives. It propagates error signals backward, enabling parameter updates that align outputs with targets.

43. Explain how overfitting appears in training curves. **Answer:** Overfitting manifests as decreasing training loss paired with increasing validation loss. This divergence signals that the network memorizes noise or irrelevant details rather than learning generalizable patterns.

44. Describe how autoencoders compress information. **Answer:** Autoencoders learn a compact representation by mapping data to a lower-dimensional latent space and reconstructing it. The model retains only essential structure while discarding redundancy, effectively performing nonlinear dimensionality reduction.

45. Explain the meaning of latent representations. **Answer:** Latent representations encode the most informative aspects of data in a compressed form. They capture underlying structure and support tasks like visualization, clustering, and transfer learning.

46. Discuss why denoising autoencoders inject noise. **Answer:** Adding noise encourages the model to learn stable features that reconstruct clean inputs even from corrupted versions. This forces the autoencoder to identify meaningful structure rather than merely memorize input details.

47. Compare reconstruction error in autoencoders and VAEs. **Answer:** Autoencoders focus solely on reconstruction fidelity, while VAEs balance reconstruction with a regularization term enforcing latent distributional structure. This yields smoother latent spaces at the cost of slightly higher reconstruction error.

48. Discuss gradient descent in non-convex neural networks. **Answer:** Gradient descent navigates highly non-convex landscapes where saddle points and flat regions abound. Despite this, it often finds good solutions because many minima are equivalent in large models, and stochasticity helps escape poor regions.

49. Explain how dropout influences learned representations. **Answer:** Dropout randomly removes neurons during training, forcing the network to distribute information more broadly. This reduces co-adaptation and acts as a powerful regularizer that improves generalization.

50. Describe how neural networks perform feature learning. **Answer:** Neural networks learn features automatically by adjusting weights to emphasize patterns useful for prediction. Early layers detect simple structures, while deeper layers capture abstractions. This hierarchical feature extraction replaces manual engineering.